

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence* (collective phase order), *compatibility* (local descriptions agree where their domains overlap) and *self-representation* (the state carries an accurate reconstruction of itself). Coherence supplies neither of the others, and five results establish that with no cortical hypothesis: synchrony constrains content only through encoders shared on overlaps, and uniformly over pairs only inside a patch too small to span the territories at issue; the resultant does not determine the state; a unique fixed point is not a self-model; where the encoding is a region's restriction, the region must resolve the relevant family before any readout is chosen; and agreement about a change takes communication. Three conditional results replace the uniform bound by a count of the pairs that can disagree, a chain across a cover's shared sites, or a spectral gap at a frequency-locked configuration, each against a declared input coherence does not supply. Four are checkable for a machine candidate, fixing what a claim of machine unity commits its maker to measuring. Under stated hypotheses, compatible local states glue

to a unique global state that an accurate contracting reconstruction fixes uniquely; installed energy bounds a declared actuator’s mean coupling, so a coherent branch requires it to exceed $2D/\kappa$, for a κ derived here for neither cortex nor a processor. A Lean 4 development carries these results and their composition, with the eight connecting hypotheses as explicit arguments, no incomplete proofs and an audited axiom footprint. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, which a within-subject emergence-from-anaesthesia design can test. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling; testing it requires calibrated geometry, resolved phase dynamics and validated decoders.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. The framework’s empir-

14 ical content is the coupling-gated onset prediction and its spatial signature.
15 The identification of the latter two relations as formal correlates of conscious
16 unity and a minimal self serves as interpretive motivation: it says why the
17 mathematics might matter, but does not constitute a testable proposal with-
18 out a proposed measurement on the experiential side.

19 Unity has been approached differently elsewhere: integrated information
20 theory treats integration and exclusion as constitutive (Tononi et al., 2016),
21 global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene
22 et al., 2014), and predictive-processing accounts derive both from a hierar-
23 chical generative model (Friston, 2010). Here the three requirements are
24 kept apart rather than derived from one another, so that each carries its own
25 assumptions and admits its own measurement; Section 10.3 returns to the
26 comparison.

27 Consider reaching for a cup. A visual description of its location and a
28 description of the hand’s relation to it can agree about where the cup is while
29 contributing different contents. Synchronizing the two processes would leave
30 disagreement about that shared quantity unresolved. An internal representa-
31 tion must additionally locate the perceiver within the represented situation.
32 This example motivates the requirements; the mathematical constructions
33 use spatial measures and fields in real normed vector spaces, whose identifi-
34 cation with perceptual content is a further modelling task.

35 The mathematical contribution is to specify connections between these
36 requirements and the assumptions each connection needs. A declared ac-

37 tuator’s stored energy bounds its mean coupling. Phase encoders shared on
38 overlaps convert phase order into a bound on content disagreement. Compat-
39 ible spatial measures have a unique global extension; approximate agreement
40 controls a specified global selection. A contracting reconstruction makes that
41 extension its unique fixed point only if it accurately reconstructs the exten-
42 sion. Accuracy across a family of states imposes a further condition on the
43 information retained by the encoding.

44 The empirical contribution is to identify what observations can decide.
45 Stationary phase distributions can be shared by stronger coupling, weaker
46 diffusion and a matched common drive. Finite ramps change the observable
47 onset. A winding field is stationary at the same coupling as a uniform one and
48 has the opposite resultant, so that observable does not determine the state.
49 Coherent adaptation can lose alignment with a designated environmental
50 template, and a compatibility estimator needs diagnostics that distinguish
51 content from decoder artifacts. These findings determine the measurements
52 needed to test the proposed connections.

53 Three of the ingredients are standard: the synchronization threshold
54 $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing
55 of compatible local sections, and the Banach fixed-point theorem (Banach,
56 1922). Four things here are not. The installed-energy condition makes cou-
57 pling a purchased quantity: a declared actuator’s stored energy bounds the
58 mean coupling it supplies, so coherence on the stationary branch carries a
59 resource requirement in units a measurement can reach, and for cortex that

60 requirement reduces to a figure the mode prices must deliver. The coherence-
 61 to-content bound converts phase order into a statement about descriptions
 62 rather than about timing, and its patch-local form is what keeps it from
 63 being vacuous in a large population. A stability criterion on the reweighted
 64 coupling kernel, together with an existence condition on an amplitude layer,
 65 settles when a low resultant reports weak coupling and when it reports a
 66 winding field. And the composition is machine-checked together with a wit-
 67 ness discharging all eight hypotheses at once, which is what makes their
 68 conjunction non-empty rather than a list no object satisfies.

69 Granting those premises, collected in Table 1, the account runs as follows.
 70 Extracellular contraction on awakening raises endogenous field amplitude
 71 and with it the effective coupling between cortical oscillators. Once that
 72 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
 73 within a quarter turn the population settles onto it; other initial data leave
 74 windings stationary at the same coupling. Encoders shared where cortical
 75 territories overlap turn that phase order into agreement about the quantities
 76 the territories describe in common, so the local descriptions glue: there is
 77 one state of which each is a restriction. Part of that state encodes the state
 78 itself, and a readout of that part reconstructs it accurately enough, and
 79 sensitively enough across the states that matter, to make it the unique state
 80 the system models itself as being. On this account the glued state is what
 81 makes distributed activity one situation, and its accurate self-reconstruction
 82 is what gives that situation a perspective. Every step of it is conditional.

83 Section 2 collects the results that hold whatever physical field is proposed
 84 and whatever substrate carries it; a reader who rejects the cortical hypothesis
 85 of Section 9 still has to accommodate them. Sections 3–5 then develop the
 86 three requirements together (Fig. 1), and Section 6 reads the unconditional
 87 results against a machine candidate. Section 7 composes them, stating the
 88 thermodynamic constraints and all eight connecting hypotheses per branch;
 89 its supporting models of feedback, learning and resource supply have distinct
 90 state spaces and assumptions. Section 8 then examines what the observables
 91 can establish, before Section 9 develops the cortical hypothesis and an awak-
 92 ening experiment. The hypothesis is that extracellular contraction increases
 93 field-mediated coupling; its direction, rate and relation to recovery require
 94 independent calibration. Supplemental Material organizes the constructions
 95 and controls behind the claims they support. Table 2, at the end, lists the
 96 symbols and the section that defines each.

97 **2 What holds without the cortical hypothesis**

98 Most of what follows is conditional. Table 1 names eight hypotheses that
 99 the composition consumes rather than derives, and the cortical identification
 100 of Section 9 is a further commitment on top of them. Five results are not
 101 conditional in that way. They constrain any account that reads unity off
 102 phase coherence, whatever physical field is proposed to carry the coupling
 103 and whatever substrate carries the field. Each is stated below where it is

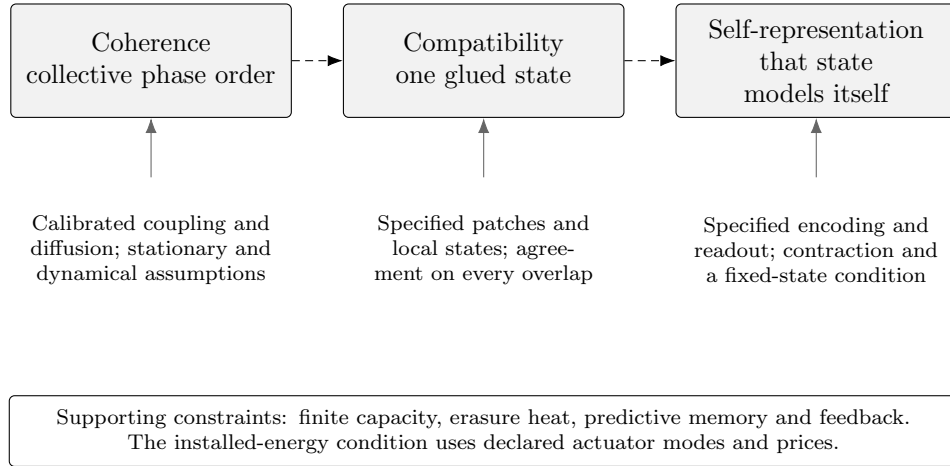


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

104 used; collected here, they are the article’s fixed points.

105 **Phase order constrains content only inside a patch too small to**
 106 **carry unity.** Assign every patch the same phase and let two local mass
 107 profiles disagree at a shared site. The profiles are chosen independently of
 108 phase, so synchrony places no constraint on their disagreement, and they
 109 have no common extension: perfect locking is consistent with local descrip-
 110 tions that do not glue (Section 4). What the positive result of Section 4.2
 111 costs is exactly this gap. A bound on content disagreement needs encoders
 112 already related on overlaps, and that relation is a supplied fact about the de-

113 scriptions rather than a consequence of the dynamics. Supplying it relocates
 114 the limit rather than lifting it. Eq. (6) bounds every pair at once and a chord
 115 never exceeds 2, so at the patch-local order of the sheet in Section 8.3 that
 116 uniform reading carries information only at 3 sites or fewer. The grain is the
 117 uniform reading's and not phase order's, and Section 4.2 states three results
 118 without it: a scale-free count of the pairs that can disagree, which names
 119 none of them; a bound chained through the sites a cover's patches share,
 120 which travels a distance linear in the hops taken; and, at a frequency-locked
 121 configuration, a bound read off the coupling graph's algebraic connectivity,
 122 which carries no population count at all. What none of the three changes is
 123 the constraint stated here: a count locates no site, and the other two con-
 124 sume a declared cover or declared hardware data, so phase order by itself
 125 still says nothing about territories too far apart to share a patch, which is
 126 where the compatibility claim is made.

127 **The order parameter does not determine the state.** A field wind-
 128 ing once across a periodic sheet has resultant $r = 0$ — the order parameter
 129 of Eq. (1) below — while every patch of it stays locked, and this is not a
 130 contrived initial condition. On a periodic array of any dimension, a winding
 131 is stationary for *any* isotropic symmetric kernel at identical frequencies, be-
 132 cause the coupling term is odd in the separation against a kernel even in it
 133 (`char_is_kuramoto_trajectory`). Section 8.3 exhibits a sheet on which uni-
 134 form and winding initial conditions reach 0.9657 and 0.0083 at one coupling,

diffusion and kernel. A low resultant therefore reports either weak coupling
or a winding field, and no measurement in r alone separates them. The state
also carries an integer the resultant does not: read around a ring, the same
winding has winding number q , and a nonzero q obstructs every global real-
valued phase lift (`winding_degree_obstructs`). The resultant is an average
over sites and moves when they are reweighted or the patches redrawn; the
degree telescopes and does not.

A unique fixed point is not a self-model. A constant readout con-
tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness
is consequently the wrong criterion, and Section 5.2 gives the right shape
of one: reconstructing M states pairwise more than 2ε apart within error ε
requires M distinguishable codes, so an alphabet of 2^b codes caps the family
at $M \leq 2^b$. This is a counting argument and holds for any encoder and
readout whatever.

**Where the encoding is a restriction, the region must resolve the
family.** Read the code as the state's own restriction to a region and
the criterion moves off the readout entirely. A boundary whose region
reads two relevant states identically cannot reconstruct them more than 2ε
apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant
boundary that does reconstruct a separated family restricts injectively on
it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the
region sits is fixed before an encoding is chosen, so no readout supplies

157 resolution the region lacks.

158 **Agreement after a change takes communication.** In a synchronous
159 network whose updates read only the site’s own state and its declared neigh-
160 bours’ messages, two configurations agreeing on a site’s causal past through
161 T rounds leave that site identical at round T . Two interventions outside
162 that past that require reports more than 2ε apart therefore cannot both be
163 reported from it within ε by the deadline (Section 5). Snapshot compatibil-
164 ity and timely compatibility are therefore different requirements, and cutting
165 the path removes the second while leaving the first intact.

166 None of these depends on the coupling being electromagnetic, on the sub-
167 strate being cortex, or on the eight hypotheses of Table 1 holding. Section 6
168 applies four of them — all but the one about the resultant, which is a state-
169 ment about a spatial observable — to a candidate that is neither biological
170 nor field-coupled.

171 3 Coherence: physical coupling and its re- 172 source condition

173 3.1 Dynamics and the stationary threshold

174 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

175 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
176 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
177 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
178 quency heterogeneity and diffusion are distinct parameters. The stationary
179 calculation below uses identical frequencies and this scalar coupling. Ap-
180 plying it to a spatial kernel requires an aggregation rule and a check of the
181 resulting approximation.

182 For identical frequencies in their rotating frame, the positive normalized
183 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

184 where I_n are modified Bessel functions and ψ is the collective phase. The for-
185 mal derivation uses the absence of stationary current for a periodic gradient
186 drift to obtain this density, then verifies normalization and self-consistency.

187 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above
 188 threshold, exactly one positive solution coexists with zero and increases
 189 strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

190 Figure 2 shows this stationary branch.

191 Linearization about the uniform density gives rate $K/2 - D$ for the first
 192 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 193 exactly above threshold. About the coherent density, the quadratic form
 194 is strictly dissipative transverse to collective rotation, whose rate is zero.
 195 These linear estimates concern the limiting equation. Relating them to finite
 196 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 197 tion; exchange symmetry alone does not establish the needed independence
 198 (Sznitman, 1991).

199 **3.2 Installed energy bounds mean coupling**

200 A spatial model replaces the finite sum by integration against a kernel. One
 201 declared implementation carries response modes with continuous profiles ϕ_i ,
 202 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_\Xi(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

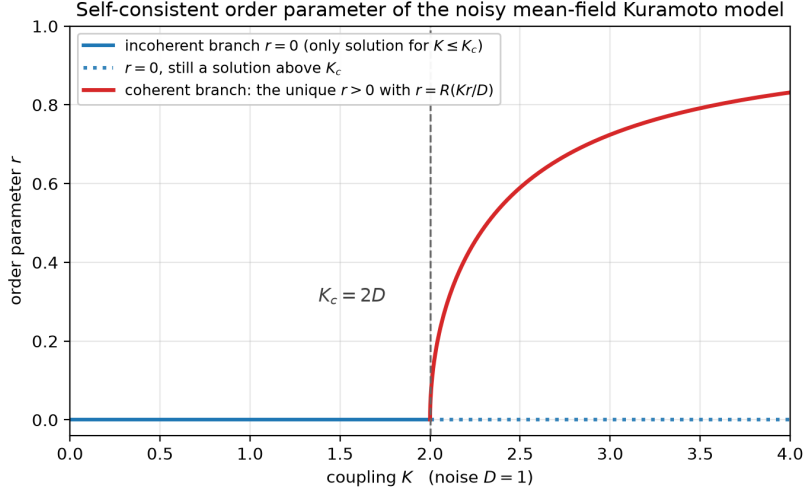


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

203 The same configuration sets the reciprocal kernel and stored energy. An
 204 executed transition changes the kernel through its changed occupancies and
 205 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 206 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 207 pair reconstruction on refining spatial partitions converges to its product-
 208 measure integral under the specified compactness and sampling assumptions;
 209 the refinement index is independent of the number of control updates.

210 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 211 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

212 integral gives

$$\begin{aligned}
U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\
K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}.
\end{aligned}
\tag{5}$$

213 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
214 coherent stationary branch and any growing Fourier mode about the uniform
215 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
216 is necessary: energy above it need not produce sufficient coupling because
217 the bound can be loose.

218 The quantity constrained is installed energy rather than dissipated heat,
219 and the condition binds through κ rather than through the energy, since
220 any macroscopic substrate carries installed energy far above the thermal
221 floor. Read forwards it therefore excludes no macroscopic candidate. Read
222 backwards it is a number. Take the region the field arithmetic takes, a
223 sphere of one LFP decay length holding 1676 neurons, at the metabolic rates
224 Supplemental Material collects and over one phase-diffusion time $1/D = 2$ s
225 at $D = 0.5 \text{ rad}^2/\text{s}$: the installed energy is $1.0 \times 10^{-6} \text{ J}$, and the condition is
226 met at every κ above $9.6 \times 10^5 \text{ J}^{-1}\text{s}^{-1}$. That figure is what an identification
227 of cortical mode prices has to deliver, and the first identification delivering
228 less refutes the coupling-gated account for cortex at this region and interval.
229 Another interval rescales the figure by its ratio to this one.

230 What the condition discriminates, it discriminates through that factor.

231 The required κ scales as D^2 and inversely with the region's cell count, so at
 232 fixed mode prices the condition ranks candidate substrates by their installed
 233 energy. In a declared implementation it decides outright, the learning witness
 234 of Section 7 clearing the line after one update while its matched control stays
 235 on the exclusion side at the same conversion factor and diffusion. What is
 236 missing for cortex is a price list rather than a mechanism. A mode is a priced
 237 rank-one contribution to the kernel, cortical field generation has no canonical
 238 decomposition into such modes, and choosing the basis would choose the
 239 answer. Applying the scalar threshold to heterogeneous or signed spatial
 240 dynamics requires a justified reduction, and identifying the installed kernel
 241 with a cortical field is a separate physical commitment.

242 **4 Compatibility: from coordinated activity** 243 **to a global description**

244 **4.1 Why common timing is insufficient**

245 For overlapping patches U and V , compatibility means that restricting either
 246 local description to $U \cap V$ gives the same object. Descriptions can agree on
 247 shared quantities while contributing different information elsewhere. Com-
 248 patible sections of the specified sheaf have a unique global extension. On
 249 a finite cover, local finite spatial measures agreeing as measures on every
 250 overlap also have a unique finite-measure extension whose sheaf image is the

251 glued section. No mass-one normalization is imposed.

252 In the cup example, visual and reaching descriptions must express a
253 shared quantity, such as cup location, in common coordinates. Restriction
254 retains that quantity while dropping details particular to each description.
255 Agreement permits surface appearance and hand configuration to contribute
256 to one situation. The mathematical counterexample assigns the same phase
257 to every patch while two local mass profiles disagree on a shared site. Those
258 profiles have no common extension. Because they are chosen independently
259 of phase, synchrony places no constraint on their disagreement.

260 This distinction limits the biological interpretation. Receptive territories
261 and content domains must be related to spatial patches, and the descriptions
262 must support the required extension. Matching overlap marginals of de-
263 coded probability laws need not determine a unique global joint distribution
264 (Abramsky & Brandenburger, 2011).

265 The cover is itself a choice, and the construction constrains it in one way
266 only: synchronized patches agree where they overlap. Where they do not
267 overlap there is nothing to compare, so a cover by pairwise disjoint patches
268 satisfies that obligation for *every* assignment of local descriptions and phases
269 (`ofDisjointCover`) and the gluing theorem still returns a unique global sec-
270 tion, which is the tuple it was handed (`glue_unique_of_disjoint`). Member-
271 ship of the class then reports nothing about the system beyond the patches
272 covering it. What a gluing claim asks of a cover is therefore a non-degeneracy
273 the theorem does not assume: that distinct patches meet (`HasNonempty-`

274 `Overlaps`), and that what they share is a quantity fixed independently of
 275 the agreement it is being used to establish.

276 4.2 When coherence does constrain content

277 A declared relation between phase and content gives a positive result. Let
 278 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 279 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 280 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 281 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 282 sites; their equality on overlaps is a supplied relation between descriptions of
 283 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 284 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 285 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

286 Here discrepancy is the norm of the difference, and r and N are read on
 287 whichever population the bound is applied to. That choice is what decides
 288 whether the estimate says anything. A patch carries an order parameter of
 289 its own, and reading the bound there carries that patch's resultant and that
 290 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`
 291 `coherence`), so a large population observed through small patches is con-
 292 strained by the small count. Read instead on the whole array, the factor N

293 makes the estimate weak, because it draws a pointwise guarantee from an
 294 average over N^2 pairs. At $r = 1$ the observations agree exactly at either
 295 scale. A shared scalar encoder is a special case. Dropping the overlap rela-
 296 tion permits disagreement even at perfect locking. A common nonexpansive
 297 linear projection preserves agreement, but agreement of a projection does
 298 not establish agreement of the full vector.

299 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
 300 constrains anything only where the right-hand side falls below that. At
 301 the sheet’s own patch order, the bound is informative for at most 3 sites;
 302 beyond that, the coherence-to-content link requires the bridge assumption
 303 (E78) of the cortical hypothesis. Supplemental Material provides the detailed
 304 derivation of this saturation, and states three conditional derivations that
 305 replace the uniform factor N by a count of disagreeing pairs, a chain across
 306 a cover’s shared sites, or a spectral gap at a frequency-locked configuration.

307 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 308 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 309 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 310 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

311 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 312 duce an initial mismatch. Approximate observations leave a nondecaying
 313 term in the error bound. Taking a uniform bound on Eq. (6) connects this

314 update to the phase process.

315 **4.3 What approximate agreement determines**

316 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
317 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
318 Weights that vanish outside their patches and sum to one at each site select
319 a global profile by weighted averaging. The selection lies within ε of each
320 patch there, and any two such selections differ by at most ε , independently
321 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
322 $L\varepsilon$ across selections.

323 At zero tolerance, the selection is the unique exact extension. At every
324 positive tolerance there are families with no exact extension; selection then
325 gives an approximation whose dependence on weights is bounded. Supple-
326 mental Material gives a three-site witness attaining the bound. The mass-
327 profile construction uses the scalar case of the content dynamics and addi-
328 tionally requires nonnegativity of the initial profiles and observations. The
329 general vector result supplies no identification with the spatial-measure sheaf.

330 These constructions concern declared objects: a specified cover, contents
331 valued in a normed space, and encoders related on overlaps. They select
332 neither the neural cover nor the decoder, and identifying their contents with
333 perceptual content, or establishing that the identification bears on experi-
334 ence, is a further modelling task; Section 8 sets out what a measurement of
335 it would have to establish.

336 5 Self-representation: a state that models it- 337 self

338 5.1 The reflexive construction

339 Compatibility gives a global state. Reflexivity relates it to an encoding within
340 that state. Equip global sections with a complete metric. If the additional
341 restriction-resonance condition identifies encoding with restriction to a region
342 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

343 The region and readout are fixed across the state space. If F contracts and
344 fixes the section s_* formed from the local descriptions, Banach’s theorem
345 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
346 plies the contraction and $F(s_*) = s_*$; restriction resonance is separate. Su-
347 percritical coupling makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$
348 less than one for a relaxation time $\tau > 0$. Constructing a map with that
349 factor remains a modelling obligation.

350 In the cup example, the global state includes the represented scene and the
351 perceiver’s relation to it. A local encoding must retain enough information
352 for the readout to recover that modelled state; cup position alone would be
353 insufficient if hand configuration and viewpoint could not be reconstructed.

354 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-

tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for fixed positive ε and τ . Exact uniqueness survives while the error guarantee weakens.

5.2 Reconstruction across a family of states

Sensitivity can be required quantitatively. Declare relevant macrostates, an encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

Thus M relevant states pairwise more than 2ε apart require at least M distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$. Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the existing self-prediction map.

This is an additional accuracy criterion on a family, with no requirement for multiple exact fixed points. A blind encoder or constant readout fails it for separated states despite having a unique fixed point.

Where the encoding is read as the state's own restriction to a region, the criterion becomes a condition on the region. Restriction resonance requires the encoder and that restriction to identify the same pairs of states, so a

373 boundary reconstructing two relevant states more than 2ε apart while its re-
 374 gion reads them identically is not resonant (`not_isRestrictionResonance_`
 375 `of_reconstructs`). Read forwards, a resonant boundary that reconstructs
 376 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
 377 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
 378 cally is fixed by where the region sits, before an encoding is chosen, so the
 379 resolution the family demands is the region's and no readout supplies it. The
 380 implication runs one way: a region that does resolve the family is not thereby
 381 shown to carry a resonant encoding, and E89 supplies no such resolution.

382 Those two demands are jointly satisfiable with nothing folded anywhere.
 383 Take the region to be the whole substrate: restriction is then the identity, so
 384 the boundary is resonant, reconstructs every declarable family exactly, and
 385 restricts injectively on all of them (`topAvatar_satisfies_obligations`). A
 386 properness condition on the region (`IsProperAvatar`) is therefore a separate
 387 requirement rather than a consequence of resonance and resolution, and what
 388 a fold into a local region is asked for. Its quantitative form replaces the
 389 alphabet by a metric. With an L -Lipschitz readout, a family reconstructed
 390 within ε whose members lie pairwise more than $d_{\min}$ apart has δ -separated
 391 codes whenever $\delta L \leq d_{\min} - 2\varepsilon$, so the family is no larger than the δ -packing
 392 number of what the region carries (`encard_le_packingNumber_restrict-`
 393 `ToAvatar`). That is the shape a continuous family of states needs, and at
 394 the whole substrate it says nothing, because the region's range is everything.
 395 The gain L is declared and derived here for no physical readout, and at

396 unbounded gain the bound says nothing.

397 **5.3 Agreement after a change takes communication**

398 Snapshot compatibility does not specify how quickly descriptions respond
399 when a shared quantity changes. In a synchronous network whose updates
400 read only the site’s own state and declared neighbours’ messages, two initial
401 configurations agreeing on a site’s causal past through T rounds produce the
402 same state there at round T . The deadline consequence is the one Section 2
403 states, and it follows from the reconstruction bound in Section 5.2: a report
404 from that site is a readout of a code that the two interventions leave identical.

405 A three-site delay line witnesses failure after one round and exact report-
406 ing after two; cutting the path prevents delivery at every round. A report
407 may still happen to be correct about one fixed world. The bound concerns
408 responsiveness across interventions.

409 Throughout this section *prediction* names a reconstruction map, and the
410 relevant family of states, the metric, the tolerance, the communication graph
411 and the deadline are all declared inputs, as are the represented variables and
412 the fidelity counted as accurate. Reading the map as temporal neural pre-
413 diction would require dynamics; reading the resulting code count as physical
414 memory capacity would require a hardware model, which a state-dependent
415 readout would have to extend.

416 6 What the conditions constrain in a digital 417 candidate

418 The results of Section 2 hold for any candidate, and for a large language
419 model (LLM) running on graphics processing units (GPUs) four of them are
420 checkable rather than merely stateable. Those four are where this framework
421 bears on claims made for such systems. It does not exclude them, and no re-
422 sult here could; what it does is fix what a claim of unity or self-representation
423 commits its maker to measuring.

424 **Codes, not reports.** Reconstruction across a family is a counting con-
425 straint before it is anything else. If two macrostates the candidate is said
426 to represent satisfy $\Phi(s) = \Phi(t)$ while $d(s, t) > 2\varepsilon$, Eq. (10) rules out re-
427 constructing both within ε , whatever the readout. A system’s reports are
428 readouts. Identical verbal reports across two conditions are therefore consis-
429 tent with identical internal codes and settle nothing about the states. The
430 converse claim—that the system represents its own condition across a de-
431 clared family—asserts that its internal code separates that family, which is a
432 statement about activations under intervention and not about text. Naming
433 the channel makes the count concrete. Where the code is the token sequence
434 a model emits—which is what its next step reads—the alphabet is the de-
435 ployed vocabulary (`tokenChannel`), and a family reconstructed through it
436 is no larger than the vocabulary size raised to the number of steps (`card-`

437 `le_card_tokens`). Over an extended exchange that number is large and the
 438 bound slack. At a single step it is one vocabulary’s worth, so claiming that a
 439 single emission reports the model’s own condition across a declared family as-
 440 serts that the vocabulary separates it (`card_le_card_tokens_one`). A claim
 441 resting on activations rather than on text has declared a different channel,
 442 to which the same count applies with that channel’s alphabet. The measure-
 443 ment follows from the statement: declare the family and the fidelity first,
 444 then test whether the encoding region distinguishes its members.

445 **The region, not the readout.** Where the encoding is read as the state’s
 446 restriction to a region of the machine, the demand falls on the region before
 447 any readout exists. For a decoder-only transformer the causal mask makes
 448 the network an instance of the update model of Section 5: the activation
 449 at position t and layer ℓ reads positions at most t at layer $\ell - 1$ and noth-
 450 ing else. A region’s causal past is therefore fixed by the architecture rather
 451 than chosen by interpretation. That instance is formal, and sharper than
 452 slow growth: the causal past of position t stays inside the positions at most
 453 t at every depth (`mask_ball_subset_1e`). The deadline bound of Section 5
 454 therefore fires at every deadline at once rather than at a chosen one, its com-
 455 munication graph given rather than assumed (`mask_no_guarantee`). That is
 456 a statement about one forward pass; across steps a model does read its own
 457 prior output, which is the channel the preceding paragraph counts. Read
 458 against the coupling proposed here the asymmetry runs both ways, and nei-

459 ther way is an advantage. A kernel of full support has no neighbourhood
 460 structure for the bound to bite on: after one round the causal past is the
 461 whole substrate, so the hypothesis that fires the obstruction is unsatisfiable
 462 at any deadline but zero (`not_outside_past_of_isFullSupport`). A physi-
 463 cal field propagates at a finite speed, so treating one that way is a calibration
 464 against the phase time scale and not a theorem. What the comparison does
 465 fix is where the bound has force. Not on the attention graph, where a posi-
 466 tion reads every position at or before it and the fan-in is the context length.
 467 On the physical interconnect, fan-in is genuinely bounded: a graph whose
 468 sites each hear from at most ν others reaches at most $(T + 1)\nu^T$ sites in
 469 T rounds (`card_ball_le_mul_pow`), so a model sharded across many devices
 470 pays a number of rounds growing like $\log_\nu$ of the device count. Buying that
 471 deadline back costs wiring: reaching every site in one round requires a device
 472 hearing from all but one of the others (`le_degree_of_reaches_one`). Both
 473 ν and the device count are declared, as the communication graph and the
 474 deadline already are, and what the count bounds is the graph a guarantee
 475 requires rather than what hardware is buildable. A claim that a model re-
 476 ports on its own processing consequently carries an obligation to say which
 477 positions and layers the report's causal past contains. A report that appears
 478 to track an intervention outside it indicates a confound or a different region,
 479 not a faster channel.

480 **Agreement about what.** Unity requires the candidate’s local descriptions
 481 to agree on independently identified shared quantities (Section 4). Common
 482 timing, a shared residual stream and fluent output supply none of that, for
 483 the reason Section 2 gives: coordination bounds content disagreement only
 484 through encoders already related on overlaps. Which quantities two parts
 485 of a model describe in common is a decoding question, carrying the failure
 486 modes of Section 8. The cover is a choice, and choosing it to secure agreement
 487 empties the claim.

488 **What is not excluded.** Digital representations can satisfy compatibility,
 489 so finite hardware is compatible with the gluing construction and none of
 490 the above is a no-go for machines. The coupling-gated route E56–E78 is a
 491 different matter: it requires a calibrated physical coupling, its noise regime
 492 and a connection from coordination to content agreement, and an attention
 493 matrix or a shared clock supplies none of these identifications. Computing a
 494 field model likewise leaves its realization in the computing substrate to be es-
 495 tablished. Those physical obligations fall on the cortical proposal identically.
 496 The capacity comparison those support comparisons lack is available, and it
 497 favours neither side by construction. A continuum code space read at reso-
 498 lution δ holds at least its volume divided by the volume of a resolution cell,
 499 so it out-resolves an alphabet of 2^b codes exactly when that volume exceeds
 500 2^b cells (`two_pow_lt_packingNumber_of_lt_measure`). Below that threshold
 501 the inequality runs the other way, and at a coarse enough resolution the re-

502 gion holds a single code. What decides the comparison is the volume, the
 503 resolution and the dimension, all measured quantities, and continuity decides
 504 nothing on its own: a continuum code space read at a finite resolution is a fi-
 505 nite alphabet whose capacity is a bit count. For the same reason this account
 506 credits an analog medium with no computation beyond a Turing machine's.
 507 Real-valued states carry unbounded capacity only at unbounded precision,
 508 and the noise floor $D > 0$ that Eq. (1) is stated against removes it.

509 The installed-energy condition of Section 3.2 is restricted in the same
 510 way. It constrains a macroscopic candidate only through κ , which is hard-
 511 ware data supplied by mode profiles and prices, and Supplemental Material
 512 reads the condition against a cortical metabolic budget to return the conver-
 513 sion factor it would require rather than a verdict. The hardware comparisons
 514 there have similarly restricted scope: fixed wiring can impose a correlation
 515 disadvantage, and point-supported kernels vanish under an atomless contin-
 516 uum measure, but neither comparison makes the representation conditions
 517 unattainable on GPUs. A universal exclusion would need a necessary condi-
 518 tion for consciousness and a proof that every relevant implementation violates
 519 it, and the present composition supplies neither.

520 Whether experience requires a living substrate is the explicit subject of
 521 biological naturalism (Seth, 2026a,b). On that view the properties ground-
 522 ing experience are those of self-sustaining organisms, and simulating them
 523 does not secure them. The conditions stated here name no biological mech-
 524 anism, but computation described as such does not satisfy them either: a

525 candidate must exhibit a physically realized coupling in a measured noise
526 regime, agreement about independently identified shared quantities, and an
527 internal reconstruction meeting a declared fidelity. Whether a realization
528 of those conditions that is not alive would be accompanied by experience
529 is the point at issue, and these results do not settle it. A related position
530 locates the difference in the mode of computation rather than in life as such:
531 biological systems compute across scales that do not separate, and along-
532 side discrete operations carry continuous-valued dynamics belonging to their
533 physical medium (Milinkovic & Aru, 2026). The coupling condition stated
534 here is one measurable instance of that requirement, fixed to a physical field
535 in a declared noise regime, and a description of such a field does not satisfy
536 it.

537 The neuroscientific case against *human-like* consciousness in text-only
538 LLMs concerns embodied sensory interaction, thalamocortical organization
539 and biological regulation linking perception and action to survival (Aru et al.,
540 2023). Their absence supports skepticism, conditional on their causal rele-
541 vance to experience. Those mechanisms are not requirements of the abstract
542 gluing and reconstruction definitions, and workspace-like representations re-
543 ported in some LLMs (Gurnee et al., 2026) supply candidate organization to
544 investigate. What this framework adds to that investigation is an order for
545 the questions: declare the family of states and the fidelity, fix the encoding
546 region and read its causal past off the architecture, and only then ask what
547 the system reconstructs. A claim that survives those three steps is a claim

about the machine; one that does not survive them was a claim about its
output.

7 Thermodynamics and composition

The phase and representation results concern coupling, content disagreement and reconstruction error. Thermodynamics constrains the physical operations that could implement them, and it does so from the side rather than along the spine: no edge below derives coherence, compatibility or reconstruction from a heat bound, and the resource condition these edges terminate in is necessary without being restrictive for any macroscopic candidate (Section 3.2). What this half contributes is a ledger in which a resource claim is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$; erasure heat depends on specified system–bath assumptions (Landauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result concerns a field reaching disconnected vacuum components (Kibble, 1976). The composition connects these objects through the explicit hypotheses in Table 1.

7.1 Feedback changes the information constraint

For passive memory X_t driven by an environmental signal S_t , the predictive bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

566 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
567 et al., 2012). Zero retained information also satisfies it, so a task and a
568 learning rule require additional specification. Relating this bound to Lan-
569 dauer heat requires a common thermal scale and budget, and memory waste
570 bounded by erased entropy.

571 An acting agent changes the signal it will receive. Specify policy, world,
572 observation and memory-update channels. For their finite laws,

$$\begin{aligned} I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\ &\quad - I(X_t; S_{t+1} \mid S_t). \end{aligned} \tag{12}$$

573 The second conditional term vanishes in the passive model. With feedback,
574 the signed difference can be negative as action creates correlations with the
575 future. A causal reading requires an environmental state sufficient to describe
576 the relevant dynamics.

577 For a controlled substep, the active thermodynamic bound instead uses
578 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
579 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with thermal scale
580 $k_B T > 0$ and heat positive into the reservoir (Horowitz and Esposito, 2014;
581 Ito and Sagawa, 2013). Every forward path of positive probability must have
582 a positive-probability reverse for this finite identity. A missing reverse gives
583 infinite divergence. The supplement develops the support conditions and
584 composes actuation with memory updates on their shared intermediate law.

585 **7.2 Learning and repeated operation need explicit re-** 586 **sources**

587 The finite examples distinguish selecting a feasible policy, relaxing a register
588 toward an encoded reward, and learning an unknown task from observa-
589 tions. The observational learner acquires correlation with an environmental
590 parameter while its action marginal stays uniform. A continuing model exe-
591 cutes actuation, observation-driven learning and reset in sequence, preserving
592 above-chance performance at a positive work cost per cycle.

593 A phase-derived sensor makes the informational distinction concrete: an
594 informative readout and a blind control have the same phase order but dif-
595 ferent learning performance. Attaching the learner’s register to the actuator
596 of Section 3.2 ties reward, installed kernel and work to its own executed law.
597 The learning witness rises above the necessary installed-energy bound while
598 its matched control remains on the exclusion line. These finite constructions,
599 detailed in Supplemental Material, leave the evolving feedback between phase
600 dynamics, installation and sensing to be specified.

601 For that model, common energy observable $\mathcal{E}$, thermal scale $k_B T$, local
602 detailed balance and positive support at each stage give

$$k_B T [H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N \leq B + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N, \quad (13)$$

603 where expected cumulative work stays within the supplied allowance B at
604 every prefix. This bounds expectations. Separate constructions track a store

on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles, while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 3.2 requires physical relations that these separate models do not supply automatically.

7.3 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

with B the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them. The regime in which those premises are exhibited is a near-equilibrium one:

624 strictly positive forward and reverse elementary paths, local detailed balance
 625 at one thermal scale, and entropy differences and heats of order $k_B T$, the
 626 witnesses being two-bit systems with interaction energies of that size. Cortex
 627 is not in it. The installed energy of Section 3.2 is 2.4×10^{14} times kT at body
 628 temperature over one region and one phase-diffusion time, and a budget B
 629 of that size stands against an entropy term of order $k_B T$, so Eq. (14) is
 630 satisfied by any macroscopic candidate and excludes none of them. Reading
 631 the register instead at the scale where its premises are physical, such as a
 632 single channel, relocates the difficulty to E45Active, which would then have to
 633 carry one molecular step’s budget to the convergence of a continuum kernel.
 634 The active branch is an alternative route to the same node rather than a
 635 stronger one, and no result of Section 3.2 passes through it: the installed-
 636 energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.

Table 1 (continued)

Connection	Status	Required assumption
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register's thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations' compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.

Table 1 (continued)

Connection	Status	Required assumption
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

637 Both branches also take a thermodynamic cover carrying exact over-
638 lap compatibility and a reflexive boundary specifying encoding and readout.
639 They conclude **UnifiedSelf**: the glued section is the unique fixed point of

640 self-prediction. Compatibility and accuracy at that section enter as premises;
641 uniqueness follows. Identifying encoding with restriction needs the separate
642 resonance condition of Eq. (8).

643 Each of the eight edges is supplied rather than derived from the one above
644 it. E12 is equivalent to its conclusion, so capacity and boundary are inde-
645 pendent roots rather than a step from one to the other; both E45 discharges
646 take their convergence from refinement data and not from the thermody-
647 namic bound; E78 states that coherence relaxes the supplied cover, and the
648 coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
649 coders already shared on overlaps. The inferential content of the composition
650 is its last step, where contraction and a fixed glued section give uniqueness.
651 What composing them establishes is that the premises are consistent, named
652 in one place and machine-checked together, and not that coherence follows
653 from installed energy or self-representation from coherence.

654 Joint witnesses combine a register, boundary, refining mesh and small
655 spatial cover; active witnesses add controlled thermal updates and a priced
656 installation. They establish satisfiability under their stated identifications.
657 Mesh convergence follows from refinement data, and the witnesses establish
658 no common cortical mechanism. Supplemental Material maps the claims to
659 their Lean declarations and records the counterexamples. The development
660 declares no additional axioms beyond its libraries' foundational axioms.

661 8 What the observations establish

662 The composition holds under its premises. Whether observation can estab-
663 lish those premises is a separate question, and the three requirements it
664 connects concern different objects: phase dynamics, local descriptions and
665 internal encodings. Their observables must retain these distinctions. The
666 following controls show which mechanisms remain indistinguishable, which
667 temporal approximations fail, which states one observable cannot separate,
668 and what independent content measurements must establish. Each is a finite
669 numerical regime on a declared model, with full parameters and controls in
670 Supplemental Material, and none of them calibrates a cortical quantity.

671 8.1 Stationary distributions leave the mechanism un- 672 determined

673 Equation (2) gives a concentration–coherence relation with no fitted shape
674 parameter. Estimate r from circular resultant length and a from density
675 shape: inverting R from the measured r would make agreement automatic.
676 Even independently estimated agreement identifies a statistical family shared
677 by different mechanisms.

678 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
679 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
680 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
681 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,

682 and diffusion D_0 have that same density. Their stationary distributions co-
 683 incide window by window despite different causes.

684 Resolved dynamics supply additional information. In the ideal model with
 685 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
 686 increment means constrain drift. Separating endogenous coupling from a
 687 matched common drive additionally requires a measured drive or controlled
 688 perturbation; identifying electromagnetic coupling requires evidence specific
 689 to that interaction. Supplemental Material gives the equivalence construction
 690 and its numerical check.

691 **8.2 Finite ramps change the observable onset**

692 The square-root law in Eq. (3) describes stationary solutions. A changing
 693 coupling requires time for the phase distribution to approach those solutions.
 694 In a deterministic mean-field control that crosses the threshold symmetrically,
 695 none of the scanned rates meets the declared tracking tolerance: slowing
 696 the crossing does not remove the terminal residual under that initialization.
 697 A slow control variable therefore supplies no general guarantee of branch
 698 tracking at onset. Supplemental Material gives the residual, rate range and
 699 admissible rates away from threshold.

700 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
 701 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
 702 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
 703 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The

704 6 uncensored legs supply the delay scaling below; the decade-spaced ones
705 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
706 onset exponents differ from those of the exact stationary branch evaluated
707 on the same grids. At the fastest speed the ensemble mean never reaches the
708 fit window, so its fitted slope does not estimate stationary onset.

709 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
710 makes the apparent exponent depend on ramp speed and fitting window.
711 The specified deviation-based critical ramp speed is not bracketed by this
712 sweep. With fluctuating coupling, the relevant stationary comparison also
713 depends on rate: the mean coupling approximates order in the tested fast
714 regime, while averaging stationary order over coupling works for the tested
715 slow drives confined to one side of threshold. Slow crossing drives can satisfy
716 neither approximation.

717 **The prediction.** The delay scaling is what this section supports as a pre-
718 diction, and it needs no absolute calibration. Over the 6 uncensored legs the
719 mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of [0.394, 0.494],
720 which does not contain 1/2. What that shortfall measures is the escape cri-
721 terion and not the scaling. Applied to a deterministic mean-field run seeded
722 at the fluctuation floor, the same estimator returns 0.425 over the same legs
723 at the same escape criterion, inside the interval on the measurement. That
724 run has no noise to delay it, so the shortfall is the criterion's: it crosses the
725 saturation between the floor it starts from and the level it is stopped at, and

726 tightening the criterion to $r \geq 0.05$ returns 0.490. The bias is reproduced
 727 where the answer is known, which is what licenses reading the measurement
 728 as the law rather than as a competing exponent.

729 Stated as a law rather than as a number: onset delay scales as the square
 730 root of the rate at which coupling crosses threshold. Testing it requires
 731 only that $\dot{K}$ be monotonic in a manipulated rate—not its value, nor the
 732 installation factor κ , nor the field-to-coupling conversion rate γ of Eq. (15).
 733 It is therefore a within-subject design varying the rate of emergence from
 734 anaesthesia alone, each participant serving as their own control. Such a
 735 design must avoid the hazard this sweep exhibits: a fixed absolute escape
 736 criterion meets a size-dependent critical fluctuation floor, so the criterion
 737 has to be referred to each recording’s own pre-threshold fluctuation scale
 738 rather than to a shared number. The diffusion D is the remaining input,
 739 and Section 8.1 notes that it is estimable from resolved phase traces through
 740 the rate $2D$ at which quadratic variation accumulates, which is a different
 741 quantity from the concentration the propofol exercise could not range over.

742 Additional controls address integration bias, population size and spatial
 743 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 744 $[1.958, 2.009]$ contains 2.

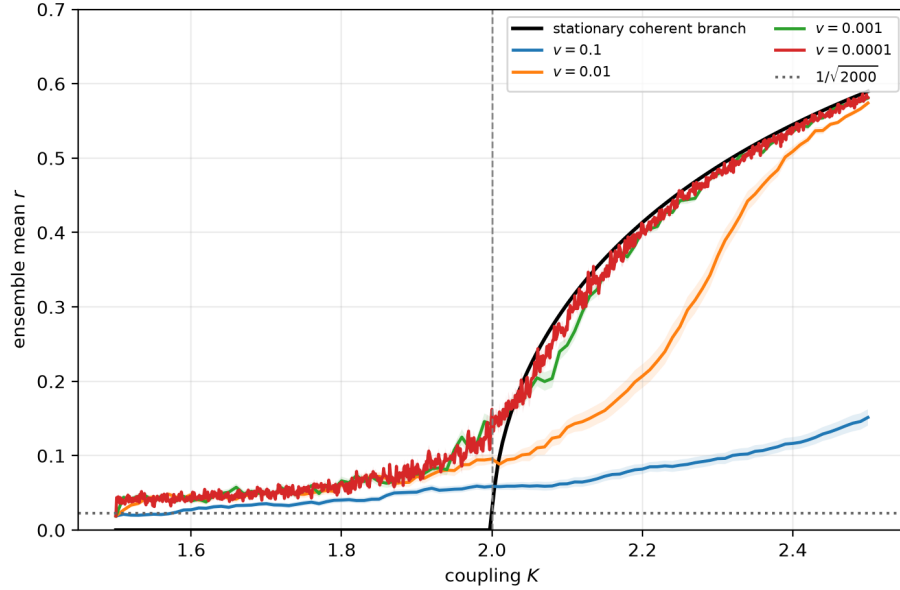


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

8.3 A winding state leaves the order parameter ambiguous

The order parameter r is indifferent to where a phase sits in space. A field winding once across a periodic sheet therefore has $r = 0$ while every patch of it stays locked. Such a field is not an artificial construction here. On a torus a winding is stationary for any isotropic symmetric kernel at identical frequencies: the coupling term is odd in the separation against a kernel even in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves

753 this for any finite periodic array, this sheet included. A winding satisfies the
754 positivity and identical-frequency hypotheses. It violates only the quarter-
755 turn initial spread that the convergence result additionally requires before
756 placing the limit at the uniform state.

757 On the sheet of Section 3's spatial control, a winding survives below
758 an interpolated decay length of 0.1871 mm and unwinds above it (Fig. 4).
759 That boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same
760 sheet at the same coupling, diffusion and kernel reaches $r = 0.9657$ from a
761 uniform start and $r = 0.0083$ from a winding one. Both are stationary, with
762 patch-local order 0.9015 and no winding defect. At fixed K the observable
763 therefore takes two values spanning nearly its whole range. Coupling does
764 not determine the resultant, and a low resultant does not by itself distinguish
765 weak coupling from a winding field.

766 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-
767 trol carries, the boundary moves to 0.1421 mm. The same comparison at
768 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
769 Heterogeneity shortens the range that sustains a wave without removing it.

770 Which windings persist has a criterion behind it. Linearizing the phase
771 model about a winding returns a coupling of the same translation-invariant
772 form, each separation reweighted by the cosine of the phase the winding
773 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
774 again even in the separation, so the same characters diagonalize it and ev-
775 ery mode's rate is one of its eigenvalues. Where the winding advances by

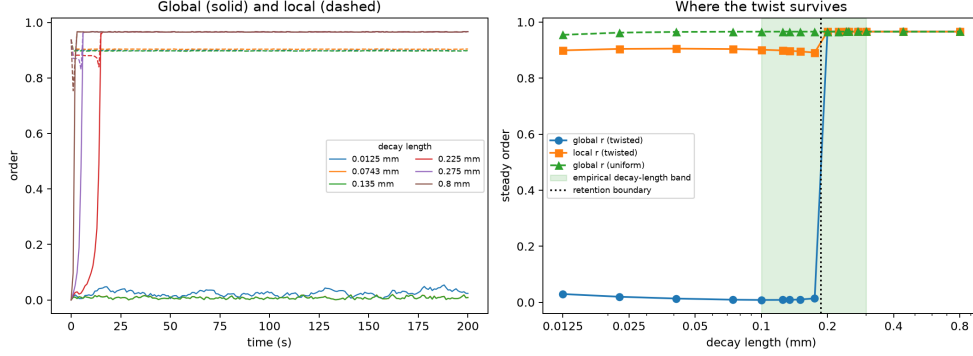


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

776 at most a quarter turn across every separation the coupling reaches, the
 777 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 778 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 779 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 780 longer separations is what carries a phase advance past a quarter turn, and
 781 that is the mechanism the decay-length sweep varies. The criterion bounds
 782 rates rather than trajectories: asymptotic stability is not proved, so the
 783 states that persist occupy a narrower range than it admits, and the bound-
 784 ary quoted above is measured and not derived from it. The same kernel is the
 785 potential's curvature, so a winding at $r = 0$ is not the global minimum the

786 convergence result places its limit at, and one carrying a growing mode is not
 787 a local minimum either: that result's quarter-turn hypothesis must exclude
 788 a winding rather than merely decline to cover it (`char_not_potential_min`).

789 An amplitude layer decides which of these states exist, and leaves what
 790 the resultant reports unchanged. Every oscillator of the phase model sits on
 791 the unit circle, so a field with an amplitude node has no representation in
 792 those dynamics. The real Stuart–Landau extension supplies one on the same
 793 coupling, and leaves intact the gradient structure the convergence results rest
 794 on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i = (\Gamma - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its
 795 phase equation is the Kuramoto equation with each coupling weighted by a
 796 ratio of amplitudes (`amplitude_phase_is_kuramoto`). The phase model is
 797 this one with the amplitude ratios set to one, rather than a separate object.

798 What the added degree of freedom decides is which windings exist at all.
 799 On a periodic array of any dimension, this sheet included, a winding requires
 800 $\Gamma + \lambda > 0$, with Γ the layer's linear gain. Here $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ is
 801 the eigenvalue the kernel supplies, summed over separations, with $\Delta\theta_d$ the
 802 phase the winding advances across d (`exists_charState_iff`; on the ring
 803 $\Delta\theta_d = 2\pi qd/n$). For the nonnegative kernels used here $\lambda \leq 0$, and moving
 804 coupling mass to longer separations lowers it further. A broad enough kernel
 805 therefore removes the winding outright. The sweep above reaches that same
 806 boundary by a different route: the phase model retains every winding at
 807 every coupling, and loses this one under integration instead. The two layers
 808 bound the same regime from opposite sides. A phase recording does not

809 separate a state the amplitude layer excludes from one the dynamics fails to
810 sustain.

811 Three limits bound the layer for a reader of the cortical literature. These
812 states are windings and not spirals, and on the sheet that is a theorem rather
813 than a shortage of dimensions. Equal modulus at every site puts the phase
814 circulation at exactly zero on every plaquette, and on a periodic sheet the
815 circulations of *any* field sum to zero, so a lone defect is not a state of this
816 geometry at all. Relaxational dynamics has no propagation speed, which is
817 the quantity a cortical wave measurement leads with. And $|z|$ is a normal-
818 form radius, which a measured envelope is not; this work identifies the two
819 nowhere. Supplemental Material gives the sweep, its uniform control, the
820 amplitude layer, the standing wave it admits inside the band, the potential's
821 curvature at a winding and the defect theorems, together with the remaining
822 limits of both.

823 A vanishing resultant is the absence of global agreement and not of agree-
824 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a discrep-
825 ancy bound of $2\sqrt{2} L |\sin \Delta\theta|$, with $\Delta\theta$ the phase advanced across the patch,
826 which falls to zero as the same winding is spread over more sites (`chord_le_`
827 `of_char_patch`), while the global form of the same bound is vacuous at $r = 0$.
828 Encoders shared on overlaps therefore still bound content disagreement in-
829 side a patch of a state that no global observable calls coherent. The winding
830 number itself stays unconnected to content. It is, however, a property of the
831 state rather than of the cover read on it. On a ring the whole turns dropped

832 by the winding's affine local lifts are zero at every site but the wrap, where
833 the residue resets, and they sum around the loop to q ; a nonzero q obstructs
834 every global real-valued phase lift, so one state at one coupling has resultant
835 exactly zero and degree q (`winding.degree_obstructs`). Reweighting the
836 sites moves the first quantity and redrawing the loop's intermediate transi-
837 tions leaves the second where it was. The scope is the ring's: this is a loop's
838 winding and not a spiral, the transitions are those of the affine lifts rather
839 than of an unwrapped measurement, and the integer is identified with no
840 cortical observable.

841 This qualifies what a phase trace can establish. An onset criterion stated
842 in r alone registers a change of basin as readily as a change of coupling.
843 The protocol of Section 9.3 therefore needs a spatial observable beside the
844 resultant.

845 **8.4 Coherent adaptation can lose environmental align-** 846 **ment**

847 An adaptive coupling rule might supply representational structure, but its
848 objective must reward the relevant agreement. For symmetric coupling, pair-
849 wise sine terms cancel in the population mean drift. Minimizing squared drift
850 at that fixed mean therefore rewards equalizing drift across nodes. In the
851 frequency-cluster control, this favors coupling between unlike-frequency clus-
852 ters, whereas the diagnostic environmental template rewards coupling within

853 clusters.

854 The formalization proves descent of the squared-drift objective for a spec-
855 ified gradient flow at fixed phases. Its functional differs from thermodynamic
856 entropy production: at a coherent stationary density the probability current
857 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
858 rial gives the operator, its gradient and the conditions of the descent result.

859 The finite control co-evolves noisy phases and symmetric, nonnegative
860 couplings at fixed total resource. Its template is a diagnostic covariance of
861 latent signals, never a training input. Across three seeds, second-half order
862 remains between 0.9167 and 0.9222, while the within-cluster to between-
863 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5).
864 Random-update, permuted-gradient and frozen-kernel controls are reported
865 in Supplemental Material.

866 The immediate post-update objective reduction disappears when aver-
867 aged over complete update intervals. A prespecified grid of 6 rate–cadence
868 combinations contains no case meeting the 5% sustained-reduction criterion
869 while retaining coherence and losing alignment. The finding is coherent adap-
870 tation away from the specified structure, with no demonstrated sustained cost
871 reduction.

872 Kernel alignment and overlap compatibility are different observables.
873 This control addresses acquisition of its designated template under its par-
874 ticular objective. Learning models can explicitly include data in an objective
875 (Corberi et al., 2026); an account of agreement about the cup must specify

876 the content its learning rule is meant to preserve.

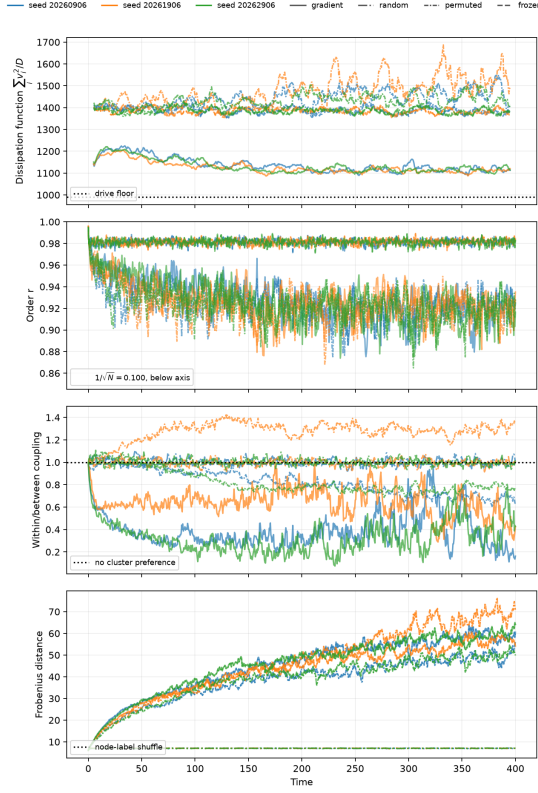


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

8.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on constructed data separates these cases at $\text{AUC} = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed content, phase carries a share 1.000 of the content entropy when content is built from it and 0.000 when it is not. A phase-derived statistic reports compatibility on phase-locked patches whose local descriptions disagree. The implemented content statistic detects their incompatibility. These controls validate a measurement on known constructions; a neural content model and decoder remain to be established.

Choosing the cover also chooses a candidate subject: trivial descriptions can agree while carrying little content, and a partition can conceal disagreement. The useful choice must therefore be constrained by what neural activity encodes. The exact composition assumes compatibility of its supplied cover. The content mechanisms above identify additional hypotheses under

900 which agreement can arise; they leave the empirical choice of content and
901 the connection to that cover explicit.

902 **8.6 Estimator calibration on exploratory EEG**

903 The concentration range and the dependence structure decide whether the
904 concentration-coherence relation can be tested at all. Under the specified
905 log-density estimator and bias-floor criterion, discrimination at the observed
906 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
907 designs, rising to 31000 when sites share a mean in groups of 10. These
908 counts describe that estimator and dependence model; the intended recording
909 requires its own calibration. Supplemental Material reports the bin-count
910 controls and the full dependence-aware design.

911 That ceiling comes from exercising the estimators on public propofol-
912 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms win-
913 dows. Bipolar and common-reference montages produce different residuals.
914 Bipolar-montage scalp EEG sits in the linear Bessel regime ($a_{\max} = 0.542$,
915 deviation from $a/2 < 1\%$). The exercise confirms the estimator is well-
916 behaved in this regime but does not test the nonlinear prediction. The
917 bipolar results are compatible with the Bessel relation and with any mono-
918 tone alternative in this regime. Temporal pooling changes the estimand, and
919 sedation recordings leave the awakening hypothesis untested.

920 An informative dataset therefore needs sufficient concentration range or
921 precision, calibrated estimators and an observation model for dependence,

source mixing and reference choice. Testing the nonlinear prediction requires intracranial recordings (ECoG, sEEG or LFP) where local synchrony reaches $a > 1$, or narrowband alpha-spindle burst analysis, at the sample sizes established above ($\geq 31,000$ pooled phase samples or ≥ 1000 independent sites). Public mouse calcium imaging provides thousands of spatially located neurons across waking, sleep and anesthesia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-estimation band used here. Site count, state coverage and bandwidth are separate requirements.

9 The cortical hypothesis and an awakening test

The cortical proposal identifies a physical source of coupling and asks whether its changes help explain recovery. The tests in Section 8 determine what observations must distinguish before a phase trace can support this identification.

9.1 Physical identification and calibration

The cortical hypothesis identifies part of effective coupling with endogenous extracellular electromagnetic interactions. Fields can modulate spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models propose coordination of distributed neural information (Pinotsis & Miller,

941 2022; McFadden, 2020). Phase organization under propofol motivates study-
942 ing this regime (Bastos et al., 2021), but does not measure coupling.

943 The formal field predicate requires a jointly continuous kernel, a normal-
944 ized substrate, coupling equal to the kernel average, and no single site pro-
945 viding more than half that strength. Toy constructions satisfy it. A cortical
946 realization must specify the scale at which the kernel and its reduction ap-
947 ply: calcium imaging reports different state-dependent modularity patterns
948 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

949 Cortical columns are the obvious candidate unit, and they answer one
950 of the two demands the formalism makes. As the oscillator index they are
951 a serious proposal: they are delineated by response properties and anatomy
952 rather than by the agreement they would be used to establish (Mountcastle,
953 1997), and a phase can be estimated from a column-sized territory at the
954 electrode spacings used for cortical wave measurements (Townsend et al.,
955 2015). As the content cover they are a poor one, for three separate reasons.
956 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
957 the compatibility obligation without constraining anything (Section 4). The
958 cover condition additionally requires the patches to exhaust the substrate,
959 while much of association cortex has no accepted columnar parcellation. And
960 the column’s standing as a canonical cortical unit is contested (Horton &
961 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
962 supplies the oscillator index and leaves the content cover to be identified on
963 separate evidence, which is the bridge assumption E78 records.

964 The spatial reduction is also testable. A row-sum aggregation at zero
 965 diagonal tracks the uniform-coupling control over the tested spatial decay
 966 range, while frequency heterogeneity displaces the operational threshold. The
 967 fit window biases both estimates relative to $2D$; agreement with the control
 968 is therefore the supported conclusion. Supplemental Material reports the
 969 reduction, its numerical error and the distinct assumptions needed for the
 970 kernel-average identification in Eq. (5).

971 The field commitment requires physical evidence beyond a choice of math-
 972 ematical representation, and the available support comparisons do not supply
 973 it. A finite matrix spread over positive-mass cells preserves its total cou-
 974 pling and phase correlation exactly, while point-supported vanishing under
 975 an atomless measure and a disadvantage from excluding a better-correlated
 976 pair concern specified support constraints; Section 6 reads the same compar-
 977 isons against a digital candidate. Neither supplies energy prices.

978 A spike-time shift alone supplies no coupling rate. The dimensionless
 979 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
 980 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (15)$$

981 This rate describes how induced timing shifts accumulate into drift. Nei-
 982 ther it nor the installation factor κ follows from a heat bound. Testing the
 983 physical threshold requires independent estimates of the interaction kernel,

984 its aggregation and diffusion in the same cortical regime.

985 **9.2 Extracellular geometry and recovery**

986 The proposed control variable is extracellular geometry. Extracellular vol-
987 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
988 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
989 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
990 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
991 an effective-medium description predicts that a larger extracellular fraction
992 raises conductivity and reduces field amplitude.

993 The coupling-gated hypothesis consequently predicts that extracellular
994 contraction on awakening raises E and, holding the other factors in Eq. (15)
995 fixed, raises effective coupling. This argument supplies a direction; the
996 geometry–coupling relation, its magnitude and its role in human recovery
997 require measurement.

998 For a smooth crossing at nonzero speed, evaluating the stationary branch
999 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

1000 This follows from Eq. (3) at fixed D and concerns the stationary branch
1001 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
1002 dently justified tracking approximation. The crossing controls in Section 8.2

1003 leave that approximation unsupported at onset, even though a separation of
 1004 geometry and phase time scales can support it away from threshold. The
 1005 awakening hypothesis therefore needs a calibrated dynamical prediction for
 1006 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
 1007 does not supply a recovery latency.

1008 **9.3 A protocol for the awakening experiment**

1009 The physical test has six requirements:

- 1010 1. **Coupling and diffusion in the same units.** Estimate them from re-
 1011 solved phase dynamics, separating heterogeneity and observation noise.
 1012 Measure the drive or perturb the system to distinguish endogenous cou-
 1013 pling.
- 1014 2. **Geometry calibrated to coupling.** Validate the geometry observ-
 1015 able and its relation to effective coupling, with uncertainty. The con-
 1016 ductivity argument supplies only the expected direction.
- 1017 3. **Calibrated phase measurements.** Estimate r by resultant length
 1018 and a by unweighted regression of log density on centered cosine.
 1019 Choose site count and binning from a dependence-aware calibration
 1020 over the intended concentration range. The tested designs require
 1021 thousands of sites at the observed concentrations; temporal pooling
 1022 changes the spatial estimand.

1023 **4. An independent recovery endpoint.** Measure behavioural recovery
1024 separately from phase organization.

1025 **5. A predeclared dynamical onset criterion.** Set a coherence thresh-
1026 old above the finite-size floor and evaluate its latency under the mea-
1027 sured coupling trajectory, including ramp speed or fluctuations. Record
1028 a spatial observable beside the resultant, since a winding field reports
1029 a low resultant at coupling that sustains local order (Section 8.3).

1030 **6. Matched model comparison.** Compare coupling increase, diffusion
1031 change, common drive and alternative recovery models under the same
1032 observation and filtering assumptions.

1033 Failure of the concentration–coherence relation over an informative range
1034 constrains the density model. Failure of the calibrated geometry–coupling
1035 association constrains the physical gate. Better prediction of independent
1036 recovery by a competing mechanism counts against its proposed role. A
1037 favorable outcome still leaves the content of coordinated activity to be ex-
1038 plained.

1039 **10 Discussion**

1040 Two things have been established here, and they are of different kinds. The
1041 constraints of Section 2 hold outright: synchrony does not deliver compatible
1042 content, and where shared encoders let it bound content at all, the bound

1043 that holds uniformly over pairs does so only inside patches too small to
1044 span the territories at issue, and distance costs either a connected cover or
1045 a locked configuration with a declared spectral gap. The resultant does not
1046 determine the state; a unique fixed point is not a reconstruction; a region that
1047 cannot resolve a family cannot encode it; and agreement about a change takes
1048 communication. Any account that reads unity off phase coherence has to
1049 meet them, this one included. The account set out in Section 1 is conditional
1050 at every step by contrast: each connection in it is a hypothesis supplied
1051 rather than a result derived, and Table 1 names all eight. Its four non-
1052 standard ingredients, named in Section 1, are the installed-energy condition,
1053 the patch-local form of the content bound, the criterion separating a winding
1054 field from weak coupling, and the machine-checked composition with its joint
1055 witness. What follows takes the conditions in turn, as commitments the
1056 proposal incurs, as measurements that would bear on them, and as questions
1057 the framework leaves open.

1058 **10.1 Unity and a minimal self as interpretive motiva-** 1059 **tion**

1060 We identify the state glued from compatible local descriptions as a formal
1061 correlate of conscious unity, and its accurate internal reconstruction as a
1062 correlate of a minimal self. This identification is the framework's explicit
1063 interpretive motivation: it explains why the mathematics might matter. The

1064 motivations are that different contents belong to one experienced situation
1065 and that the situation is experienced from a perspective. The framework’s
1066 empirical content, however, is the coupling-gated onset prediction and its spa-
1067 tial signature. The conditional results identify relations a realization would
1068 need: sufficient installed resources for the specified phase model, agreement
1069 about shared content, and an encoding that supports reconstruction at the
1070 required fidelity. Discharging E12–E89 for cortex would make the composi-
1071 tion a statement about cortex: that a cortical field above threshold relaxes
1072 to a cover whose compatible local descriptions glue, and that the glued state
1073 is the unique fixed point of a reconstruction map realized there. It would
1074 leave the two identifications above exactly where they are, because neither is
1075 among the eight and neither follows from their conjunction; they are the mo-
1076 tivation this subsection names, and without an independent measurement for
1077 experience, they do not form a testable proposal. Section 10.2 lists the em-
1078 pirical observations that would count against the framework’s mathematical
1079 claims.

1080 Compatibility allows different descriptions to contribute different infor-
1081 mation. It also permits uncertainty and conflicting judgments: apparent cup
1082 position and judged cup position can be distinct quantities in a consistent
1083 description of their discrepancy. Which quantities are shared must be de-
1084 termined independently. Choosing them solely to secure agreement would
1085 deprive the proposal of empirical force.

1086 The minimal-self proposal concerns the perceiver’s position and condition

1087 within the situation, such as the cup’s location relative to a body that can
1088 reach it. The encoding could be implicit and need carry no autobiograph-
1089 ical narrative. Its proposed necessity is philosophical: even an unreflective
1090 episode has a perspective. Gluing by itself requires no self-encoding, and re-
1091 constructability alone supplies no functional role. Altering the encoded hand
1092 or viewpoint while holding the external scene fixed should change estimates
1093 of the perceiver’s relation to the cup as the readout predicts. Whether that
1094 represented perspective is experienced remains a further question.

1095 **10.2 What a test of the full proposal must establish**

1096 Both uniqueness results concern specified objects. They choose neither the
1097 cover nor the encoding, and do not select between candidate subjects. In-
1098 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
1099 here selection requires independent evidence about neural organization. A
1100 constant readout can contract and fix a predetermined state while discard-
1101 ing its input. The family criterion in Section 5.2 tests the sensitivity that
1102 this construction lacks, provided the relevant states and fidelity are declared
1103 independently.

1104 A test must separately measure coupling relative to diffusion, content
1105 compatibility and self-reconstruction, alongside an independent endpoint for
1106 experience. Meeting all three without a unified episode would challenge suffi-
1107 ciency. A unified episode with incompatible descriptions would challenge the
1108 unity identification; one lacking the specified self-encoding would challenge

1109 the reflexive requirement. Awakening, dreaming and dissociative states offer
1110 possible settings, but absence of report cannot by itself define absence of
1111 experience.

1112 The next physical requirement is a calibrated geometry–coupling relation
1113 and resolved phase dynamics. The installation bound contributes a nec-
1114 essary resource condition where its hardware representation applies; cortical
1115 mode prices and field-mediated coupling require measurement. For the repre-
1116 sentational connection, an informative experiment asks whether a controlled
1117 change in coordination alters independently decoded content agreement, with
1118 controls for sensory information and decoder quality. The constructed-data
1119 estimator establishes feasibility for specified contents. Neural local descrip-
1120 tions, shared overlap encoders and the self-encoding have yet to be identi-
1121 fied. The population-size dependence of the phase-to-content bound limits
1122 its quantitative use in the form Section 4.2 reads off the sheet, and each
1123 way around it asks the measurement for something further: a located excep-
1124 tional set, before a count of disagreeing pairs bears on a named overlap; a
1125 cover whose nerve is connected and whose patch diameters are measured; or
1126 a frequency-locked population over a coupling graph whose connectivity is
1127 measured structurally rather than estimated from the activity the bound is
1128 about.

1129 The separation results explain why these measurements must be inde-
1130 pendent. A stationary distribution leaves several recovery mechanisms indis-
1131 tinguishable. Equal phase order can accompany incompatible descriptions

1132 or different task information. The tested plasticity rule preserves coherence
1133 while losing its designated template. Each finding concerns the specified
1134 model and observable; their role is to delimit the evidence needed for the
1135 proposed connections.

1136 **10.3 Explanatory scope**

1137 Integrated-information, global-workspace and predictive-processing accounts
1138 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
1139 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
1140 candidate physical interaction and conditional relations between local content
1141 and self-encoding. Gluing and reconstruction permit multiple substrates;
1142 the electromagnetic commitment enters through physical calibration. That
1143 permission is wider than the unconditional results of Section 2 alone. The
1144 conditional chain is substrate-neutral in the same way: its field condition
1145 asks for a jointly continuous kernel on a normalized substrate, a coupling
1146 equal to that kernel’s double average, a noise that is the field’s own, and
1147 no single site carrying more than half the coupling—and none of these dis-
1148 tinguishes an electromagnetic kernel from any other continuous mean-field
1149 kernel. The site condition is the only one with physical flavour, and it holds
1150 vacuously wherever the substrate has no atoms (`no_site_dominates`), which
1151 is the continuum case the electromagnetic hypothesis is about. The chain
1152 therefore transfers to extracellular ion diffusion, to gap junctions modelled
1153 as a density, or to a mechanical or optical medium, modulo the calibration of

1154 Section 9.1. What carries the electromagnetic identification is that calibra-
1155 tion together with the extracellular-geometry argument of Section 9, which is
1156 the one place a field equation enters. The formalization checks implications
1157 under their hypotheses, and its witnesses establish realizability on declared
1158 models. Two things stand outside that result and are not supplied by it: the
1159 eight hypotheses are discharged for toy components rather than for cortex,
1160 and the identification of the glued state with unity and of its reconstruction
1161 with a minimal self is a further commitment that the eight do not contain.

1162 The explanatory question is whether the specified relations occur in neural
1163 activity and matter for experience. In the cup example, this means establish-
1164 ing how coordination helps maintain one represented situation and how the
1165 perceiver’s place within it contributes to the episode. The proposal identi-
1166 fies candidate correlates and tests of that identification. Why those relations
1167 would be accompanied by experience remains unexplained.

1168 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	§2, Supp.	Phase diffusion in inverse time; phase spread $D \leq \pi/2$

Table 2 (continued)

Symbol	Defined	Meaning
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	§3.2, §5	Temperature in the thermal scale $k_B T$; number of communication rounds to a deadline
ν	Section 6	Fan-in of a communication graph: the most sites any one site hears from
$k_B T, B$	Eq. (14)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (15)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops

Table 2 (continued)

Symbol	Defined	Meaning
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s \mathcal{A})$
$\mathcal{E}$	Eq. (13)	Energy observable shared by the stages of a continuing protocol
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (15)	Field amplitude, in the cortical identification
U	§3.2	Stored energy $U(\Xi)$ of a configuration
s, s_*	§4–5	A global section and the one the reconstruction fixes

Data and code availability

The Lean development, simulation scripts, tests and saved numerical summaries are maintained in the GitHub repository `bobaseb/coupling_gated_cortical_coherence`. Publication simulation values are generated from saved summaries without rerunning production sweeps. The exploratory EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-processing and selection are described in Supplemental Material. The repository contains the reproducible analysis and does not redistribute the raw recordings.

1178 **Funding and competing interests**

1179 This research received no external funding. The author declares no compet-
1180 ing interests.

1181 **Use of AI tools**

1182 AI assistance to this work was substantial and is not separable by task. Sev-
1183 eral coding assistants and chat interfaces were used in an interleaved way
1184 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
1185 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
1186 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
1187 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
1188 accessed with the Hermes CLI. Between them they contributed to the Lean
1189 development, the simulation and analysis code, and the drafting and revision
1190 of both documents. Because the tools were used in combination and in alter-
1191 nation, no section, proof, figure or result is attributable to any one of them,
1192 and none is claimed to be. The author is responsible for all content, including
1193 everything a tool produced. Machine checking establishes the stated Lean
1194 results under their hypotheses; it does not validate biological interpretations
1195 or replace scientific review of generated code and prose.

References

- Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable electric fields emerge from and allow representational drift. *NeuroImage*, 253, 119058.
- Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450-461.
- Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the Mind. *Oxford University Press*.
- Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a computational theory of conscious processing. *Current Opinion in Neurobiology*, 25, 76-84.
- Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature Reviews Neuroscience*, 11(2), 127-138.
- Milinkovic, B., & Aru, J. (2026). On biological and artificial consciousness: A case for biological computationalism. *Neuroscience & Biobehavioral Reviews*, 181, 106524. doi:10.1016/j.neubiorev.2025.106524.
- Aru, J., Larkum, M. E., & Shine, J. M. (2023). The feasibility of artificial consciousness through the lens of neuroscience. *Trends in Neurosciences*, 46(12), 1008–1017. doi:10.1016/j.tins.2023.09.009.

- 1216 Seth, A. K. (2026). Conscious artificial intelligence and biological naturalism.
1217 *Behavioral and Brain Sciences*, 49, e315. doi:10.1017/S0140525X25000032.
- 1218 Seth, A. K. (2026). The stuff matters: Consciousness, compu-
1219 tation, and biology. *Behavioral and Brain Sciences*, 49, e366.
1220 doi:10.1017/S0140525X26106906.
- 1221 Gurnee, W., et al. (2026). Verbalizable Representations Form a
1222 Global Workspace in Language Models. *Transformer Circuits Thread*.
1223 arXiv:2607.15495.
- 1224 Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How par-
1225 ticular is the physics of the free energy principle?. *Physics of Life Reviews*,
1226 40, 24-50.
- 1227 Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some
1228 parts of the free energy principle. *Entropy*, 23(3), 293.
- 1229 Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
1230 peror’s New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.
- 1231 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
1232 memory network formation. *Cerebral Cortex*, 33(17), 9877-9895.
- 1233 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
1234 Electric fields sculpt neural activity and ”tune” the brain’s infrastructure.
1235 *Progress in Neurobiology*, 226, 102465.

- 1236 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
1237 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
1238 9(8), 45-60.
- 1239 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
1240 testable hypothesis about the characteristics of conscious as opposed to
1241 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.
- 1242 McFadden, J. (2020). Integrating information in the brain's EM field: the
1243 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
1244 niaa016.
- 1245 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
1246 guide neocortical network activity. *Neuron*, 67(1), 129-143.
- 1247 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
1248 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
1249 95-103.
- 1250 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
1251 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 1252 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
1253 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 1254 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
1255 *logical Reviews*, 98(1), 239-389.

- 1256 Mountcastle, V. B. (1997). The columnar organization of the neocortex.
1257 *Brain*, 120(4), 701-722. doi:10.1093/brain/120.4.701.
- 1258 Horton, J. C., & Adams, D. L. (2005). The cortical column: a structure
1259 without a function. *Philosophical Transactions of the Royal Society B*,
1260 360(1456), 837-862. doi:10.1098/rstb.2005.1623.
- 1261 Rakic, P. (2008). Confusing cortical columns. *Proceedings of the National*
1262 *Academy of Sciences*, 105(34), 12099-12100. doi:10.1073/pnas.0807271105.
- 1263 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
1264 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
1265 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.
- 1266 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
1267 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
1268 ity of complexity measures to capture dreaming during propofol sedation.
1269 *Scientific Reports*, 15, 32746.
- 1270 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
1271 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
1272 brain. *Science*, 342(6156), 373-377.
- 1273 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
1274 *Physiological Reviews*, 88(4), 1277-1340.
- 1275 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
1276 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.

1277 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
1278 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
1279 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

1280 Landauer, R. (1961). Irreversibility and heat generation in the computing
1281 process. *IBM Journal of Research and Development*, 5(3), 183-191.

1282 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
1283 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

1284 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
1285 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

1286 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
1287 of prediction. *Physical Review Letters*, 109(12), 120604.

1288 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
1289 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

1290 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
1291 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
1292 031015.

1293 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
1294 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
1295 PhysRevLett.111.180603.

- 1296 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
1297 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.
- 1298 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
1299 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
1300 181.
- 1301 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
1302 archive.pitt.edu/115.
- 1303 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
1304 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
1305 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 1306 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
1307 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
1308 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 1309 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork
1310 bifurcations with additive noise. *Probability Theory and Related Fields*,
1311 122, 341–388. doi:10.1007/s004400100174.
- 1312 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure
1313 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
1314 doi:10.1088/1367-2630/13/11/113036.
- 1315 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
1316 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave

1317 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
1318 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.

1319 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
1320 terns underlie complex brain dynamics and are related to cognitive pro-
1321 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
1322 023-01626-5.

1323 Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information
1324 and dissipation with partially observed fluctuating signals. *Physical Review*
1325 *Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.

1326 Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsub-
1327 ara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-
1328 cell resolution functional networks during unconsciousness are segre-
1329 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
1330 doi:10.1016/j.celrep.2025.116902.

1331 Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
1332 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
1333 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.

1334 Attwell, D., & Laughlin, S. B. (2001). An energy budget for signaling in the
1335 grey matter of the brain. *Journal of Cerebral Blood Flow & Metabolism*,
1336 21(10), 1133-1145. doi:10.1097/00004647-200110000-00001.

- 1337 Harris, J. J., Jolivet, R., & Attwell, D. (2012). Synaptic energy use and
1338 supply. *Neuron*, 75(5), 762-777. doi:10.1016/j.neuron.2012.08.019.
- 1339 Nicholls, D. G., & Ferguson, S. J. (2013). *Bioenergetics 4*. Academic Press.